LESSON 8.5

Understand Inequalities and Their Solutions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can graph the solutions of an inequality on a number line.

WHY THIS MATTERS Detective Okafor is building a timeline. The suspect was seen at the scene no earlier than 2:00 PM, meaning the arrival time t satisfies $t \geq 2$. She needs to plot this on a timeline to share with her team. How should she mark the number line?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Number line *Spanish: Recta numérica*	A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right.	A horizontal line with marks at 0, 1, 2, 3, ... showing where values fall
Open circle *Spanish: Círculo abierto*	An empty circle showing a number is NOT included.	$x > 5$: open circle at 5 means 5 itself is NOT a solution, but 5.1, 6, 7, ... are
Closed circle *Spanish: Círculo cerrado*	A filled-in circle showing a number IS included.	used for $\leq$ and $\geq$
Solution set *Spanish: Conjunto solución*	All the numbers that make the inequality true.	every number 8 or above
Inequality *Spanish: Desigualdad*	A math sentence that compares two sides with <, >, $\leq$, or $\geq$.	$x \geq 3$ is an inequality; it is graphed as a closed circle on 3 with an arrow to the right.
Boundary point *Spanish: Punto límite*	The number on the number line where the solution set starts.	the 8 in $x \geq 8$

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Graph $x < 6$ on a number line.

1. The boundary number is 6.
2. < does not include 6, so draw an open dot at 6.
3. Less than 6 means the arrow points left. Test: $5 < 6$ is true.

Answer: open dot at 6, arrow pointing left

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Graph x < 5. What number is the boundary?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- My inequality is _____ because the phrase says _____ .
- I drew a _____ dot at _____ because _____ .
- My arrow points _____ because the solutions are _____ .
- The value _____ sits under my arrow, so it _____ .

Word bank inequality | graph | number line | boundary point | closed circle | open circle | arrow | shade | solution set | at least | at most | include

WATCH OUT!

A common mistake

A common mistake in Graph Inequalities is treating ≥ and ≤ the same as > and < and drawing the wrong circle. For example, students often graph x ≥ 6 with an OPEN circle (as if it were x > 6), which wrongly excludes 6 even though 6 itself makes x ≥ 6 true. A second common mistake is choosing the right circle but shading the wrong direction

LESSON 8.5

Understand Inequalities and Their Solutions

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort by circle type: open or closed.

Groups: Open circle (not included) | Closed circle (included)

_____ $x > 4$

_____ $x < 12$

_____ $n > 0$

_____ $x \geq 7$

_____ $x \leq 3$

_____ $n \geq 15$

HINT < and > use OPEN circles because the boundary is not included.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x < 5$?

(A) Open circle at 5, shaded left

(B) Closed circle at 5, shaded left

(C) Open circle at 5, shaded right

(D) Closed circle at 5, shaded right

HINT Every inequality graph asks two questions: is 5 itself a solution (circle type), and which side holds the solutions (shading)?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x \geq 3$?

(A) Closed circle at 3, shaded right

(B) Open circle at 3, shaded right

(C) Closed circle at 3, shaded left

(D) Open circle at 3, shaded left

HINT Check the symbol: ≥ has the 'or equal' bar. What does that bar mean for the circle at 3?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x > 8$?

(A) Open circle at 8, shaded right

(B) Closed circle at 8, shaded right

(C) Open circle at 8, shaded left

(D) Closed circle at 8, shaded left

HINT Two decisions: does $x = 8$ make $x > 8$ true, and are the solutions bigger or smaller than 8?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x \leq 10$?

(A) Closed circle at 10, shaded left

(B) Open circle at 10, shaded left

(C) Closed circle at 10, shaded right

(D) Open circle at 10, shaded right

HINT Ask two things: does 10 itself satisfy $x \leq 10$, and do solutions live above or below 10?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read** Graph n ≥ 8 on the number line.
2. **Circle choice** Open circle at 8
3. **Shade direction** Shade right
4. **Final graph** Open circle at 8, shaded right

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at the symbol: does it have a small line under it, like ≤ or ≥?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide between an open circle and a closed circle when graphing?

SENTENCE STARTER I use a ___ circle when the symbol is ___ because the boundary value ___ part of the solution.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.5

Understand Inequalities and Their Solutions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each inequality to its graph description

_____	Item 1	A
_____	Item 2	B
_____	Item 3	C
_____	Item 4	D

2 **Mark and label each value on the number line.**

Read the graph description and write the inequality it represents. Then graph it.



SHOW YOUR THINKING

3 **Complete the table. Show how you found each value.**

Complete the table comparing how different symbols affect the graph.

Inequality	Circle Type	Shade Direction	Is x = boundary a solution?
x > 6			
x ≥ 6			
x ≤ 6			

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A number line shows a closed circle at 8 and is shaded to the left. Which inequality does this represent?

(A) x ≤ 8

(B) x < 8

(C) x ≥ 8

(D) x > 8

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read** The case file needs fewer than 9 clues, so $c < 9$. Graph it.
2. **Circle choice** Open circle at 9
3. **Shade direction** Shade right
4. **Final graph** Open circle at 9, shaded right

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide between an open circle and a closed circle when graphing?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 8.5

Understand Inequalities and Their Solutions

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Graph the compound constraint: the suspect's age a is at least 18 AND at most 35. Graph a ≥ 18 and a ≤ 35 on the same number line. What range of ages is possible?



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Graphing Error

1. **Inequality** $x > 6$
2. **Circle type** Closed circle at 6
3. **Shade direction** Shade right
4. **Graph** Closed circle at 6, shaded right

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A theme park has two rides. Ride A requires height $h > 48$ inches. Ride B requires height $h \geq 48$ inches. A child is exactly 48 inches tall. Which ride can they go on? Explain using number line graphs for both inequalities.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How do you decide between an open circle and a closed circle when graphing?

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How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help